

KOHLER AI SALES & SPATIAL DESIGN ADVISOR

Comprehensive System Mental Model, Architectural Reference & Technical Specification

Version: 2.4.0 Production | Architecture: 4-Engine Deterministic Pipeline | Target: Luxury Spatial Commerce

1. Executive Summary & Problem Statement

Selecting and purchasing luxury bathroom fixtures involves complex multi-dimensional constraints. Traditional generative AI approaches fail in this domain due to three systemic vulnerabilities: **(1) Spatial Hallucinations:** recommending oversized fixtures into compact powder rooms that violate residential building codes; **(2) Multi-Step Arithmetic Failures:** inaccurate discount rollups and tax computations; and **(3) API Latency & Cloud Rate Limits (HTTP 429):** resulting in conversational stalls and degraded customer experience.

The **Kohler AI Advisor** solves this by decoupling generative natural language from spatial and financial logic. The system treats generative LLMs strictly as conversational polishers, delegating all spatial clearances, budget allocations, package discounts, and catalog lookups to four deterministic, verified computation engines.

2. The Mental Model: Four Deterministic Engines

The architecture operates on a strict four-layer pipeline where queries are filtered and routed at **0ms overhead** before reaching the orchestrator. If cloud LLMs lag or fail, a zero-lag deterministic fallback guarantees instantaneous proposal delivery.

Engine Layer	Primary Role & Mechanism	Latency / Cost	Key Output Deliverable
1. Traffic Controller (0ms Intent Router)	Regex pattern classifier bypasses LLM tokens for casual greetings, catalog inquiries, and single-spec fast paths.	0ms 0 Tokens	Direct Welcome Card, Portfolio Table, or Fast-Path Routing.
2. Spatial Studio & Optimizer (Constraint Engine)	Evaluates room square footage, calculates minimum/maximum fixture counts, enforces 15" centerline and 21"-30" aisle clearances, harmonizes aesthetic themes.	< 2ms Deterministic	Complete Fixture Package, Code Compliance Matrix, Dimensional Fit.
3. Financial Engine (INR & GST Math)	Executes exact Indian Rupee commerce arithmetic: 15% package volume discounts, 18% GST calculation, and budget variance checks.	< 1ms Deterministic	Itemized Quotation Table with Subtotal, Savings, GST, and Grand Total.
4. 2-Stage Hybrid RAG (Catalog Retrieval)	ChromaDB dense BGE embedding retrieval followed by Cross-Encoder reranking (ms-marco-MiniLM-L-6-v2) with in-memory LRU cache.	15-40ms Local / Cache	Top-K Kohler SKU specifications, finish codes, dimensions, and live stock.

3. Architectural Request Flow & Orchestration

USER REQUEST / SPATIAL STUDIO INPUTS

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0ms INTENT ROUTER (router.py) — Regex Intent Classification & Fast-Path Bypass

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 Greeting / Basic → Instant Direct Sales Advisor Welcome (0ms, 0 Tokens)

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 Portfolio Inquiry → Instant Structured Kohler Catalog Overview (0ms, 0 Tokens)

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 Product Specs → Fast-Path RAG (BGE Embeddings + Cross-Encoder Reranker)

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 Suite Planning / Math / Stock → Advisor Orchestrator (agent.py)

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DETERMINISTIC TOOL DISPATCH & SYNTHESIS

1. bundle_optimizer.py: Room sizing, fixture capacity, clearance validation

2. calculator.py: 15% package savings, 18% GST calculation in INR

3. inventory.py: Regional warehouse stock & postal PIN transit days

4. agent.py (3.0s Timeout): Ollama / OpenRouter LLM Synthesis

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 On Timeout or 429 Error → Zero-Lag Deterministic Markdown Fallback (0ms)

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GRADIO DUAL-TAB USER INTERFACE (app.py) — Spatial Studio + Conversational Advisor

4. Ergonomic Clearance & Building Code Standards

The Spatial Studio and Suite Optimizer enforce rigorous residential plumbing and ergonomic guidelines. Rooms below designated square footage thresholds are automatically restricted to prevent overcrowded, non-compliant layouts.

Fixture Category	Min. Centerline Clearance	Front Aisle Clearance	Architectural Constraint / Rule
Smart Toilet / Commode	15 inches (38 cm) from centerline to sidewall	21" to 30" (53-76 cm) unobstructed	Must maintain dedicated clearance envelope; placed in dry zone.
Vanity / Basin	15 inches (38 cm) from centerline to sidewall	21" to 30" (53-76 cm) unobstructed	Positioned near entry wall; mirror lighting clearances enforced.
Walk-In / Spa Shower	30" x 30" (76x76 cm) minimum interior envelope	24" (61 cm) clear door swing area	Wet-zone containment; requires dedicated floor slope/drain.
Freestanding / Alcove Tub	30 inches (76 cm) accessible side span	21 inches (53 cm) clear apron aisle	Strictly prohibited in rooms < 35 sq ft to prevent code violations.

Room Square Footage & Fixture Sizing Rules:

- **Powder Room (< 28 sq ft / e.g. 4x5, 4x6 ft):** Fixed capacity: 2 fixtures (Compact Basin + Wall-Hung Commode). Bathtubs and walk-in showers strictly excluded.
- **Three-Quarter Bath (28 - 45 sq ft / e.g. 5x7, 6x7 ft):** Fixed capacity: 3 fixtures (Vanity + Commode + Walk-in Shower Enclosure).
- **Full Family Bath (45 - 75 sq ft / e.g. 7x8, 8x9 ft):** Standard capacity: 4 fixtures (Double/Single Vanity + Toilet + Shower + Alcove/Drop-In Tub).
- **Grand Master Suite (> 75 sq ft / e.g. 10x10, 12x12 ft):** Full capacity: 5 fixtures (Statement Freestanding Tub + Intelligent Smart Toilet + Thermostatic Shower + Double Vanity + Luxury Brassware).

5. Indian Luxury Commerce & Financial Pricing Engine

All catalog products are priced in Indian Rupees (INR) following luxury retail conventions. Pricing is normalized using the standard luxury retail ending (trailing 999):

Indian Rupee Price Normalization Formula:
Catalog Price (INR) = [Round((Base USD × 83.50) / 1000) × 1000] - 1
Itemized Quotation Mathematics: • Subtotal: Sum of individual catalog MRPs. • Package Savings (15%): Subtotal × 0.15 • Taxable Amount: Subtotal - Package Savings • GST (18%): Taxable Amount × 0.18 • Grand Total: Taxable Amount + GST Amount • Budget Variance: Target Budget - Grand Total (identifies surplus or budget deficit)

6. Hybrid LLM Architecture & Zero-Lag Fallback

To provide both maximum local portability and enterprise cloud capabilities, the system implements a dual-runtime model backed by a deterministic markdown fallback:

Runtime Mode	Engine / Model	Behavior & Fallback Logic	Use Case & Portability
Local Offline Mode	Ollama Daemon (qwen2.5:3b / deepseek-r1)	Checked via 0.5s health ping. Runs 100% offline with zero cloud API keys, zero token fees, and total data privacy.	On-premise showroom kiosks, private offline workstations.
Cloud Provider Mode	OpenRouter API (meta-llama / qwen)	Protected with a strict 3.0s timeout. Catches HTTP 429 rate limits, network outages, and socket timeouts.	Cloud web deployments, mobile users, automated CI/CD.
Zero-Lag Safety Net	Deterministic Markdown Synthesizer	Instantly compiles tool outputs into structured markdown proposal if LLM fails or exceeds 3.0s timeout.	Guarantees 100% UI uptime under all network conditions.

7. Dual-Tab User Experience & Interface Design

The web application delivers two complementary workflows in a unified pure Python Gradio interface:

Tab 1: Spatial Studio & Suite Optimizer

- **Visual Control Card:** Dimension sliders (4-16 ft), budget slider (INR 30k - 12L+) with quick preset pills (1.5L, 3.5L, 6.5L, 10L+), 5 curated design themes, and fixture checkboxes.
- **Live Solution Deliverables:** Product package table with finishes and SKU numbers, itemized INR quotation breakdown, and building code compliance check.
- **Interactive Handoff:** One-click button transfers the full spatial suite proposal into Tab 2 and prompts the AI Advisor for consultative design advice.

Tab 2: Conversational AI Advisor

- **Executive Sales Assistant:** Greets clients, answers technical plumbing questions, recommends finishes, and checks warehouse logistics.
- **Custom Dark Scrollbars:** Complete cross-browser dark scrollbars matching the luxury slate aesthetic.
- **Live Streaming Badges:** Transparent status notifications showing tool dispatch, spatial calculations, and GST math in real time.

8. Automated Verification & Benchmark Matrix

All components are verified via a 12-scenario automated test suite (test_suite.py) testing boundary conditions:

Scenario ID	Evaluation Focus & Test Case	Expected Deterministic Behavior	Status
S1 - S3	Spatial Fit (4x5 Powder Room, 8x9 Family Bath, 10x10 Master Suite)	Respects 2, 4, 5 fixture limits; enforces 15" centerline and 21" front clearances.	PASS (100%)
S4 - S5	Aesthetic Cohesion (Modern Minimalist, Classic Luxury, Japanese Zen)	Curates matching finishes (Matte Black, Brushed Bronze, Chrome, French Gold).	PASS (100%)

S6 - S8	Financial Accuracy & GST Math (15% package discount, 18% GST in INR)	Zero arithmetic hallucinations; itemized subtotal, discount, GST, and total match.	PASS (100%)
S9 - S10	Catalog Retrieval & Fast-Path Routing (Veil Toilet, Purist Faucet, Moxie)	Top-K Cross-Encoder retrieval; instant 0ms routing for single-product specs.	PASS (100%)
S11 - S12	Warehouse Logistics & Resilient Fallback (Postal PIN check, 429 rate limit)	Resolves transit days (2-5 business days); 0ms fallback on LLM timeout.	PASS (100%)